

Question 1

In packet switching with store-and-forward, a packet needs to be received in full at a switching node or router before it can be placed on an outgoing link and transmitted. In cut-through switching, as soon as the first hundred or so bits of the packet (referred to as header) are received, the outgoing link can be determined, and if it is free, transmission can start immediately on that link. Consider a linear path consisting of a source host, a destination host and $k - 1$ intermediate switches/routers, where $k > 1$ (there are k intermediate links). The source has a number of packets (N) of size P bits each. The source adds an H -bit header to each packet. All the k intermediate links have a propagation delay of T seconds and a bit rate of B bps. Neglect processing and queuing delays at the intermediate switches/routers. (4 points)

- a) If transmission of the packets starts at time = 0, at what time instant are N packets received at the destination host, if packet switching with store-and-forward is used? (Hint: To be able to figure out a general formula, you may draw a space-time graph for a value N and k , e.g., $N=4$, $k=3$. You can also refer to Module1's in-class exercises.)
- b) Repeat part (a) assuming that cut-through switching is used.

Part A Answer:

- $K-1$ intermediate switches, where $k > 1$, k is the link amounts
 - Source $\rightarrow$ Dest = k is 1
 - Source $\rightarrow$ Router $\rightarrow$ Dest = k is 2, 1 router, 2 links
- N packets, size P bits each
- Source adds H -bit header to each packet
- All k intermediate links have a propagation delay of T seconds and a bit rate of B bps
- Abstract Formula:
 - Transmission Delay + Propagation Delay for each link (k)
 - Since this is store and forward, we must do this for every packet
 - The first packet is propagation delay $\times$ each link + transmission delay $\times$ every link
 - Every subsequent packet adds 1 transmission delay as we have an overlap
 - $N = 3$, $K = 2$
 - While packet 1 is at hop 2, packet 2 is at hop 1.
 - While packet 1 is at hop 3, packet 2 is at hop 2 and packet 3 is at hop 1
- $kT + ((P+H) / B) \times (N + (k-1))$

Part B Answer:

- With cut-through switching, we start transmitting as soon as the header is received.
- Abstract Formula:

- Propagation Delay for each link (k) + header only transmission delay per router + transmission delay of data portion per packet
 - We can start transmitting the next packet after the header is processed
 - The first packet arrives at $kT + H/B + P/B$
 - Because of overlapping transmitting, we add a H/B delay at each intermediate switch ($k-1$), not at each link
 - Because the overlap occurs during the header transmission, we can just calculate the transmission delay of the data portion per packet
- $kT + (k-1)H/B + N(P/B)$

Question 2

Five equal-size datagrams (packets) belonging to the same message leave for the destination one after another. However, they travel through different paths as shown in the table. We assume that the delay for each router (including waiting and processing) is 3, 10, 20, 7, and 20 ms respectively. Assuming that the propagation speed is 2×10^8 m/s, find the order the datagrams arrive at the destination and the delay for each. Ignore any other delays in transmission. (11 points)

Datagram	Path Length	Visited Routers
1	3,200 km	1, 3, 5
2	11,700 km	1, 2, 5
3	12,200 km	1, 2, 3, 5
4	10,200 km	1, 4, 5
5	10,700 km	1, 4, 3, 5

Question 2 Answer:

- 1 - 3ms
- 2 - 10ms
- 3 - 20ms
- 4 - 7ms
- 5 - 20ms

Datagram 1:

$3 + 20 + 20 = 43\text{ms}$ (total delay from all 3 routers it visits)

Propagation Delay: $3200\text{km} / (2 \times 10^8) \text{ m/s} = 3200000 / 200000000 = 0.016\text{s} = 16\text{ms}$

$$43 + 16 = 59\text{ms}$$

Datagram 2:

$$3 + 10 + 20 = 33\text{ms}$$

$$\text{Propagation Delay: } 11700\text{km} / (2 \times 10^8) \text{ m/s} = 11700000 / 200000000 = 0.0585\text{s} = 58.5\text{ms}$$

$$33 + 58.5 = 91.5\text{ms}$$

Datagram 3:

$$3 + 10 + 20 + 20 = 53\text{ms}$$

$$\text{Propagation Delay: } 12200\text{km} / (2 \times 10^8) \text{ m/s} = 12200000 / 200000000 = 0.061\text{s} = 61\text{ms}$$

$$53 + 61 = 114\text{ms}$$

Datagram 4:

$$3 + 7 + 20 = 30\text{ms}$$

$$\text{Propagation Delay: } 10200\text{km} / (2 \times 10^8) \text{ m/s} = 10200000 / 200000000 = 0.051\text{s} = 51\text{ms}$$

$$30 + 51 = 81\text{ms}$$

Datagram 5

$$3 + 7 + 20 + 20 = 50\text{ms}$$

$$\text{Propagation Delay: } 10700\text{km} / (2 \times 10^8) \text{ m/s} = 10700000 / 200000000 = 0.0535\text{s} = 53.5\text{ms}$$

$$50 + 53.5 = 103.5\text{ms}$$

Order (least to greatest):

1, 4, 2, 5, 3

Question 3

In packet switching with store-and-forward, a packet needs to be received in full at a switching node or router before it can be placed on an outgoing link and transmitted. Consider a linear path consisting of a source host, a destination host and two intermediate routers. The source has five packets of size 5000 bits each. All links have a propagation delay of 4 msec and a bit rate of 100 kbps. Ignore processing and queuing delays at the intermediate switches/routers.

a) How long does it take to receive packet 1 at the destination host? (3 points)

b) How long does it take to receive all 5 packets at the destination host? Explain your answer. (5 points)

Part A Answer:

Propagation Delay Formula: Distance / speed

Packet Size = 5000 bits

Number of Packets = 5

Propagation Delay = 4 msec

Bit Rate = 100 kbps

For 1 packet, we need transmission delay + propagation delay, for each of the intermediate routers + 1 for the final destination link

Transmission Delay = $5000 / 100,000 = 0.05$ seconds $\times 1000 = 50\text{ms}$

Propagation Delay + Transmission Delay per link = $50\text{ms} + 4\text{ms} = 54\text{ms}$

3 links: Source $\rightarrow$ 1 $\rightarrow$ 2 $\rightarrow$ Dest.: $54\text{ms} \times 3 = \mathbf{162\text{ms}}$

Part B Answer:

The transmission delay is 50ms, so after 50ms, the first packet has arrived at the first hop. Once this packet is at the first hop, the next packet can begin transmitting to the first hop while the first packet gets transmitted to the second hop. This creates an overlap in the packets being transmitted between hops, thus the calculation at the bottom.

We know that 1 packet takes 162ms total, so if we have 5 packets total, we need to calculate how long it would take for the remaining four packets to arrive at the destination. A 50ms delay is incurred per hop as that is the transmission delay.

$162\text{ms} + 50(4) = \mathbf{362\text{ms}}$